

A Deep-Learning-Based Traffic Classification Method for 5G Aerial Computing Networks

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Abstract—With the rapid progress made in aerial computing technology and the increased popularity of fifth-generation (5G) networks, uncrewed aerial vehicles (UAVs) have been playing a crucial role in real-time data collection, processing, and transmission. However, due to the diversity in traffic generated by UAVs in various mission scenarios, there is a significant challenge posed in traffic classification. Therefore, a novel traffic classification model is proposed in this article on the basis of the spatial attention-enhanced convolutional neural network (SAE-CNN). This model proves effective in improving classification accuracy and latency, particularly in the context of various 5G services, such as enhanced mobile broadband (eMBB), ultrareliable low-latency communication (URLLC), and Internet service. Also, a 5G heterogeneous network platform is built to collect UAV-related aerial computing data, with extensive experiments performed to verify the superior performance of the SAE-CNN model compared to other state-of-the-art methods. The experimental results demonstrate that the proposed approach enables effective traffic management and classification for the application of UAV in complex 5G environments.

Index Terms—5G Network, aerial computing, deep learning, traffic identification, uncrewed aerial vehicle (UAV).

I. INTRODUCTION

WITH a rapid progress made in the development of aerial computing technology, the novel computing architectures based on uncrewed aerial vehicles (UAVs), satellites, along high-altitude platforms, have been playing an important role in data collection, processing, and transmission [1]. In addition to enhancing the mobility and flexibility of these systems, this technology also enables the real-time processing of data in large volumes through distributed computing on aerial platforms [2]. In comparison to conventional ground-based computing systems, aerial computing is more responsive to the real-time demands in complex environments, for which it has been extensively applied in such settings as remote monitoring, environmental surveillance, disaster relief operations, agricultural productions, and intelligent transportation [3], [4]. For instance, in the field of smart agriculture, UAVs equipped with cameras and sensors are deployed to monitor the status of crops in real time, while edge computing devices are employed to analyze and process the data obtained, thus providing crop farmers with timely decision-making support [5].

Under this architecture, UAVs can perform three critical roles: 1) service provider; 2) information receiver; and 3) relay node [6], as shown in Fig. 1. As service providers, UAVs can capture images, videos, and sensor data, uploading the processed information to ground servers [7]. As information receivers, UAVs can acquire data from other UAVs or ground stations to make responses accordingly [8]. As relay nodes, UAVs play a crucial role in enabling communication in areas with disruptions [9]. For example, in the scenario of disaster relief, UAVs can transmit critical data from the disaster-struck area to the command center through 5G networks, enhancing the efficiency and accuracy of disaster relief operations.

Since UAV operations heavily rely on advanced communication technologies, 5G networks play a critical role in enhancing UAV communication capabilities due to their high-speed transmission, low latency, and high reliability [10], [11]. These features are essential for handling the large volumes of data that UAVs generate in real time, which conventional communication technologies cannot adequately support. By leveraging enhanced mobile broadband (eMBB) for high-bandwidth needs, such as high-resolution video transmission

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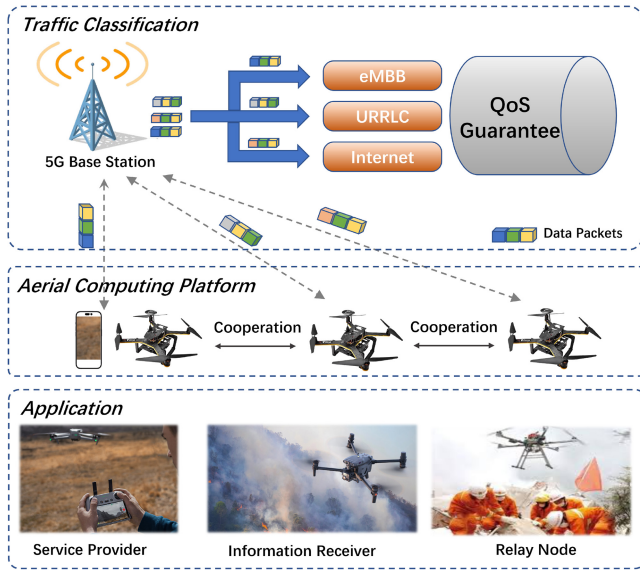


Fig. 1. Illustration of aerial computing applications and traffic classification demands in 5G networks.

and multisensor data collection in scenarios like disaster relief and environmental monitoring, 5G networks enable rapid, real-time data transmission to ground control centers [12], [13], [14], [15]. ultrareliable low-latency communication (URLLC), on the other hand, ensures low latency and high reliability for tasks requiring stability and precision, such as automated flight operations, and can facilitate UAVs as relay nodes in remote or disaster-struck areas to restore communication and support emergency services [16], [17]. While the massive machine-type communication (mMTC) is primarily intended for IoT devices with lower data requirements, eMBB and URLLC are critical for UAV missions with complex 5G service needs. However, the diverse types of network traffic generated by UAVs across various tasks necessitate efficient traffic classification mechanisms to ensure accurate, real-time identification of service-specific data, enabling optimal resource allocation and Quality of Service (QoS) across applications [18], [19], [20], [21]. Convolutional neural networks (CNNs) have been widely explored for traffic classification in 5G networks, as they automate the feature extraction process, learning patterns directly from raw traffic data and supporting efficient and scalable classification across diverse traffic types. Despite these advances, challenges remain in developing CNN-based models that can enhance classification accuracy and computational efficiency to meet the demands of high-speed, low-latency UAV operations [22], [23], [24], [25].

A variety of solutions have been proposed in the literature to address traffic classification in 5G UAV scenarios [14], [15], [20], [26], [27]. For instance, Bouzid et al. [27] highlighted the significance of UAV traffic analysis and classification, especially in the context of network slicing, which is essential for supporting distinct service requirements in 5G networks. Baltaci et al. [26] conducted an in-depth analysis of UAV traffic characteristics but focuses specifically on Control and NonPayload Communication data, thus providing insights mainly into control-level traffic between UAVs and remote controllers. Further advancing this line of research,

Chen et al. [20] explored CNN-based classification for UAV-related traffic in a 5G architecture, though it primarily centers on UAV control traffic for URLLC applications, capturing only a single dimension of UAV data. However, the diverse traffic requirements of UAVs across various tasks in 5G aerial computing remain insufficiently addressed. In summary, there is a notable lack of systematic research on traffic management and classification for UAV missions within a 5G architecture, particularly in terms of cross-service traffic classification, which is essential for fully realizing UAV applications across multiple service scenarios. To bridge this research gap, this study first establishes a 5G heterogeneous network platform to collect UAV aerial computing data, enabling real-world traffic classification. Then, a novel neural network model that supports both high-precision and low-latency traffic classification is proposed. Through convolutional layers for feature extraction and the integration of spatial attention mechanisms, the model identifies key byte positions within packet sequences, improving both classification accuracy and efficiency. The primary contributions of this paper are outlined as follows.

- 1) In this study, a 5G network is constructed on the basis of software-defined networking (SDN) technology, to support various heterogeneous communication mechanisms, including long-term evolution (LTE) and 5G nonstandalone (NSA). Three major categories of 5G aerial computing service traffic data are collected as datasets for traffic classification analysis.
- 2) A traffic classification model is proposed on the basis of spatial attention-enhanced CNN (SAE-CNN), to significantly improve the accuracy and latency of UAV traffic classification, particularly in the scenario of various 5G services.
- 3) Extensive experiments are performed to validate the SAE-CNN model, demonstrating its superior performance in classification across multiple 5G scenarios when compared to state-of-the-art (SOTA) methods. A 5G heterogeneous network platform is built to validate the proposed methods for their applicability and feasibility in the real world.

The remainder of this paper is organized as follows. Section II provides an overview of the relevant work. In Section III, the system architecture of 5G heterogeneous network is described. Section IV elaborates on the traffic classification dataset and SAE-CNN methods in aerial computing. In Section V, the experimental results are analyzed in detail. Finally, the conclusions are drawn in Section VI.

II. RELATED WORKS

A. Aerial Computing in 5G

Aerial computing has become an essential framework for supporting complex UAV missions with the advancement of 5G networks. The core features of eMBB, URLLC, and Internet services enable UAVs to transmit and process data more efficiently, enhancing performance in complex scenarios. In the field of aerial computing, real-time high-definition video acquisition and transmission are core UAV applications. Thanks to the eMBB capabilities provided by 5G networks, UAVs

efficiently capture and transmit high-resolution video in high-bandwidth environments, ensuring that critical information reaches ground control centers in real time. As analyzed by [28], 5G strategies for optimizing QoS, particularly through eMBB, are crucial for supporting smooth video transmission in scenarios like smart agriculture and disaster response. URLLC also plays a vital role in flight safety and control. The study by [29] highlights the importance of 5G URLLC technology in remote UAV control, emphasizing its low-latency benefits for real-time signal reception and status transmission, particularly in complex environments. UAVs additionally serve as communication relays within 5G networks. Yang et al. [30] explored how UAVs, acting as relay nodes, use URLLC and Internet services to extend network coverage, reliably transmitting data from remote areas to the main network. According to [14], 5G strategies focused on optimizing QoS, particularly through eMBB, are essential for seamless video transmission in applications like smart agriculture and disaster response. URLLC also plays a critical role in flight safety and control. Priya et al. [15] emphasized the importance of URLLC technology in supporting low-latency remote UAV control, which is particularly valuable for real-time signal processing and status transmission in complex, dynamic environments. Additionally, Bouzid et al. [27] examined how UAVs, when used as relay nodes within 5G networks, leverage URLLC and Internet services to extend network coverage, ensuring reliable data transmission from remote locations to the central network, even in challenging conditions. This capability supports UAVs in functioning as flexible network extenders, enhancing connectivity where conventional infrastructure may be limited. Expanding on the operational characteristics of UAVs, Baltaci et al. [26] provided an in-depth analysis of UAV traffic, specifically focusing on control and nonpayload communication traffic. This study models CNPC traffic patterns and highlights the importance of maintaining secure, robust connections, which are essential for the safe and effective management of UAV missions.

In conclusion, the technical features of 5G networks provide broad support for UAVs in the field of aerial computing, allowing them to efficiently execute a wide variety of complex tasks. To ensure stable operation and optimized performance of UAV services within 5G network environments, it is critical to classify and manage UAV traffic flows meticulously. Through such classification, network resources can be allocated more effectively, ensuring the QoS for different service types, and thereby enhancing the reliability and responsiveness of UAVs across various application scenarios.

B. Traffic Classification

Traffic classification in 5G networks increasingly focuses on machine learning and deep learning techniques to address the complexities of network traffic. Kim et al. [31] proposed an effective classification approach using recurrent neural networks (RNNs) within hybrid edge-cloud architectures, achieving high accuracy and reduced latency. Xu et al. [32] introduced TrafficGCN, a method that combines packet-level and flow-level data, utilizing graph neural networks (GNNs)

for 5G traffic classification. Several studies employ deep learning techniques for specific traffic scenarios in 5G networks. For instance, Qiao et al. [33] analyzed voice traffic in 5G using long short-term memory (LSTM) and CNN models, finding that LSTM performs best for Over-The-Top voice traffic, achieving an average accuracy of 97.68%. Raikar [34] compared the classification effectiveness of three deep learning models AlexNet, ResNet, and GoogLeNet using 5G traffic data from a campus network, concluding that ResNet achieves the highest accuracy of 95%. Recent studies have further enriched this landscape. For instance, Zhang et al. [35] proposed a lightweight approach for malware traffic classification using a broad learning (BL) architecture that significantly reduces computation costs, comparing favorably against models like ResNet and VGG in efficiency without needing GPU acceleration. Similarly, Nascita et al. [36] examined the performance and interpretability of multimodal multitask models using explainable AI (XAI) techniques, benchmarking against 1D-CNN, 2D-CNN, MLP, and LSTM to illustrate how XAI can support resource management and reliability in traffic classification tasks. Rust-Nguyen et al. [37] explored darknet traffic classification, employing CNN, CNN-LSTM, and classical machine learning techniques, such as k-NN and SVM. This work provides insights into adversarial robustness, showing how different methods respond to obfuscation attacks, with random forests (RF) demonstrating strong resilience against such adversarial scenarios.

In summary, the rapid progression in applying advanced deep learning models to traffic classification continues to advance both the precision and reliability of 5G network traffic management. These studies collectively underscore the potential of diversified model architectures to handle varied traffic types efficiently, fostering a more adaptable and secure 5G infrastructure. However, few of them have been validated considering aerial computing. To address these issues, it is proposed in this study to construct a real-world 5G test environment with UAV cooperation, as inspired by [21], which simulated various scenarios of service. With a 5G aerial computing dataset collected for training and testing, the proposed framework is evaluated against different open-source datasets like ISCXtor2016.

III. SYSTEM ARCHITECTURE

In this section, a prototype system is presented. Built-in a 5G-NSA (NonStandalone) network environment by using the OpenAir-interface (OAI) platform, it integrates SDN with open virtual switch (OVS) technologies. Supporting heterogeneous networks, the platform can be used to simulate the converged scenarios that incorporate 5G NSA, LTE, and other communication technologies. It is applied to verify the practical application of 5G networks in UAVs and other devices. The components of the network architecture and their functionalities are briefly described below. As shown in Fig. 2, the experimental platform consists of four main components: 1) the SDN platform; 2) the core network; 3) the radio access network (RAN); and 4) the terminal devices. As the following sections will delve into many technical details,

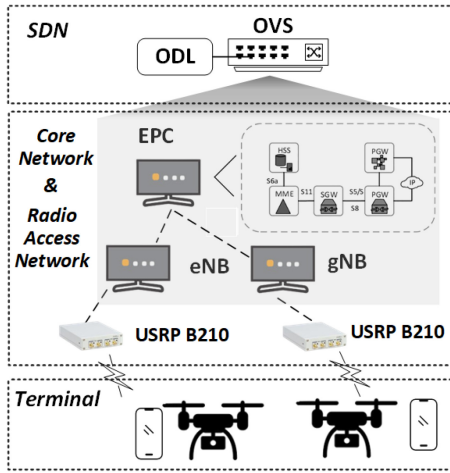


Fig. 2. Proposed 5G network system architecture.

more comprehensive information on SDN, OVS, and related technologies can be found in [38], [39], and [40]. Detailed information related to 5G technologies and architecture is available in [41] and [42].

A. SDN

1) *SDN Controller*: In this experimental platform, SDN technology is implemented using OVS and the OpenDaylight (ODL) controller to manage and configure network traffic. Through SDN, the platform is capable of dynamically controlling data flows by separating the control plane from the forwarding plane. The SDN controller is used to monitor real-time traffic in both the access and core networks. Meanwhile, network paths are dynamically adjusted according to the results of traffic classification. Thus, it is ensured that network resources are used in the optimal way.

2) *OVS*: OVS is applied to virtualize and isolate traffic. It adjusts virtual network paths in real time based on the instructions from the SDN controller, classifying different types of traffic, such as UAV control traffic and high-definition video streams. In this way, the efficient allocation of network resources is ensured when multiple terminal devices are active. Thus, the isolation and effective management of different services can be achieved in a shared physical network.

B. Core Network

1) *EPC and 5G Core Network*: The core network of the experimental platform supports both the 4G evolved packet core (EPC) and the 5G Core Network (5GC). The EPC, as the core component of the 4G network, is purposed to manage signaling, authentication, and traffic routing for 4G-connected devices while assisting the 5G gNB in dealing with user traffic. In the 5G NSA architecture, the EPC coordinates between 4G and 5G access devices, which facilitates seamless data transmission between multiple base stations for the terminal devices.

2) *Control and User Plane Separation*: The 5G core network adopts a Control and Control and User Plane Separation (CUPS) architecture, where user data are transmitted directly from the 5G gNB to the core network, with control

traffic managed by the 4G EPC. This separation improves the efficiency of data transmission in 5G networks, thus ensuring stability and reliability, particularly in case of high traffic volumes.

C. Radio Access Network

1) *4G eNB and 5G gNB*: The access network is implemented using OAI, for the simulation of an NSA environment with both 4G Evolved Node B (eNB) and 5G Next Generation Node B (gNB). The 4G eNB manages the functions performed by the control plane for devices, including signaling and connection management, while the 5G gNB focuses mainly on dealing with high-bandwidth user data traffic (user plane). In the NSA architecture, the 5G gNB relies on the signaling channel of the 4G eNB, both of which collaborate on providing data and control services to various terminal devices, such as UAVs and smartphones.

D. Terminal Devices and Traffic Types

There are various terminal devices tested in the experiment, including the UAVs equipped with 5G modules and different smartphones, such as the Huawei P40 and iPhone 12. With these devices connected to the network using different methods of wireless communication, data are transmitted over the 4G and 5G networks simulated on the OAI platform. The UAVs are used to demonstrate how the low-latency and high-bandwidth capabilities of 5G can support efficient operations in complex heterogeneous network environments. In this experiment, the UAVs equipped with 5G communication modules are connected to the network through the 5G gNB, while the 4G core network is applied to manage control plane signaling. This setup reflects the potential of 5G networks in supporting low-latency, high-bandwidth applications, particularly when real-time control and high-definition video transmission are required. By transmitting real-time data and video streams from the UAVs over the 5G network, various future applications are simulated, such as autonomous driving and industrial control.

IV. TRAFFIC CLASSIFICATION DATASET AND SAE-CNN METHOD IN AERIAL COMPUTING

Building upon the system architecture presented in the previous section, the focus now shifts to the core components required for traffic classification. This section will first describe the dataset that has been collected and used, followed by an in-depth discussion of the spatial attention-based CNN method, which forms the foundation of the traffic classification model. Fig. 3 displays the overall flowchart of the CNN-based 5G service classification experiment. First, a real-world 5G environment is created through OAI to collect 5G service data. Then, the data are preprocessed and converted into 2-D images, which are inputted into the SAE-CNN model for classification.

A. Aerial Computing Traffic Classification Dataset

To comprehensively evaluate the 5G network traffic classification model, a diverse dataset is constructed to cover a

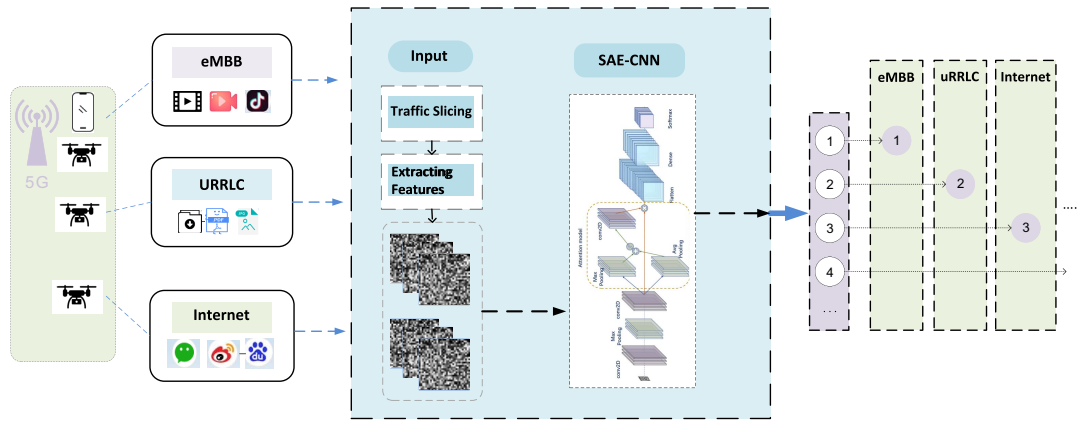


Fig. 3. Overall flow of the experiment.

TABLE I
DATASET DESCRIPTION OF VARIOUS APPLICATIONS IN 5G AERIAL COMPUTING

Terminal	Data Type	Traffic Characteristics	Label	Data Range	Scenario
Mobile Phone	Video Viewing	High bandwidth, stable transmission	eMBB	Hundreds of MB	In disaster relief or city surveillance scenarios, the user requests video recorded by UAVs from the cloud
UAV	Live Video	High bandwidth, stable transmission	eMBB	Hundreds of MB	In disaster relief or city surveillance scenarios, the UAV uploads live video feeds to the cloud
UAV	Cloud Data	Low latency, medium bandwidth, stable transmission	uRLLC	Tens of MB to GB	In UAV collaboration scenarios, the UAV uploads or downloads data to/from the cloud for processing
Mobile Phone & UAV	Video Download	Low latency, medium bandwidth	uRLLC	Hundreds of MB to GB	In disaster relief or city surveillance scenarios, users periodically receive video collected by the UAV for analysis
UAV	Social Media Information	Low bandwidth, tolerates high latency	Internet	Several MB to hundreds of MB	In disaster relief scenarios, the UAV acts as a relay node to receive or forward social media information

wide range of typical 5G network application scenarios. Packet capture (PCAP) software, Wireshark, is deployed on both 4G base stations (eNB) and 5G base stations (gNB) to generate real traffic data through public network access using UAV terminals in NSA mode. The aerial computing traffic based on various applications is listed in Table I. The focus is on input data from UAV terminals and user mobile terminals, while the computing processes of aerial computing are not considered. Additionally, the analysis targets application-layer services, excluding physical layer signal attenuation caused by the distance between UAVs and ground stations. Given the complex applications in 5G aerial computing, several representative applications are identified. Traffic classification tasks are primarily concerned with the QoS of downlink traffic [20], and the focus is placed on applications where terminal devices request services from the core network in aerial computing, as outlined in Table I. This dataset is designed to cover a broad range of real-world application scenarios, emphasizing the extensive use of UAVs in 5G networks. These applications include disaster relief, urban surveillance, real-time data uploading, and cloud interaction. The design of these scenarios addresses diverse network resource demands, showcasing the multidimensional potential of UAVs in a 5G

environment. By comprehensively recording and analyzing aspects, such as UAV data transmission and video streams, the dataset provides valuable experimental data for the aerial computing field.

1) *Overall Analysis of Packet Capture Files:* The structure of the packets captured by Wireshark is illustrated in Fig. 4. Each PCAP file contains a single file header (PCAP Header), occupying 24 bytes, and each packet within the file has a 16-byte Packet Header, which includes timestamp and data length information. The last 4 bytes of the Packet Header specify the length of the packet data. The first 36 bytes are found to be irrelevant for traffic classification network training, so only the packet length from bytes [36:40] is extracted to determine the real packet data length. Based on this, the actual packet data, including all relevant header information and payloads from the Ethernet, IP, and GTP/TCP/UDP layers, can then be extracted.

2) *Analysis of Typical 5G Data Packets:* To further illustrate the traffic characteristics of eMBB, uRLLC, and Internet applications in the 5G UAV dataset, typical service data packets from the dataset are selected for traffic feature analysis. As shown in Fig. 5, the traffic variation over time reveals that eMBB traffic exhibits significant peaks and periodic

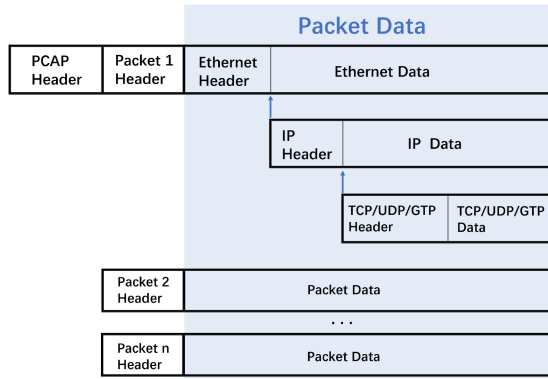


Fig. 4. Structure of PCAP packets.

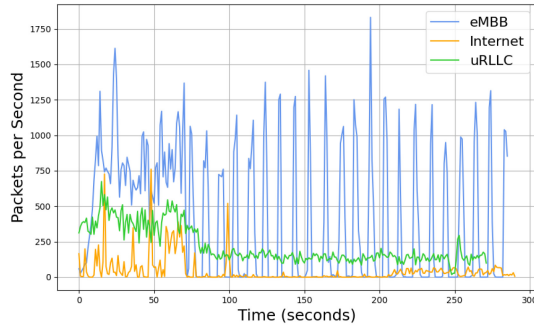


Fig. 5. Three typical categories 5G packets.

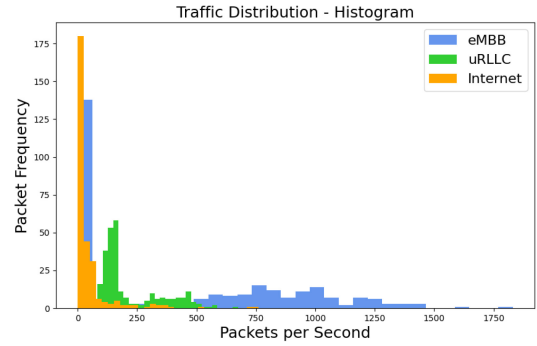


Fig. 6. Packets frequency histogram of the three typical categories 5G packets.

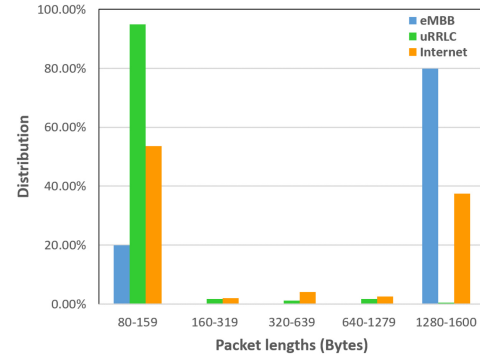


Fig. 7. Packet lengths distribution of the three typical categories 5G packets.

fluctuations, reaching up to 1500 packets per second. This substantial fluctuation reflects the high bandwidth demands of eMBB services, such as video streaming. In contrast, uRLLC traffic demonstrates relatively stable characteristics, with packet counts ranging between 200 and 300 packets per second, peaking at around 500 packets per second. This stability aligns with the low-latency, high-reliability requirements of uRLLC applications, such as remote control or real-time feedback. Internet traffic shows significant fluctuations in the first 50 s, peaking near 500 packets per second before stabilizing, indicative of bursty data transmission tasks, such as file downloads or webpage loading.

A packet frequency histogram is plotted to further confirm these observations, as shown in Fig. 6. It can be seen from the histogram that eMBB traffic frequencies are mainly distributed in the range of 500–1000 packets per second, indicating its high demands for bandwidth. Differently, uRLLC traffic is evenly distributed in the range of 100–400 packets per second, showing its stability in transmission characteristics. Internet traffic is concentrated in the range of 0–100 packets per second, reflecting its bursty nature.

3) *Packet Length Analysis*: The distribution of packet length for eMBB, uRLLC, and Internet services is further analyzed, as shown in Fig. 7. Specifically, eMBB traffic primarily features packet lengths in the 1280–1600 byte range, comprising the majority of its traffic, which indicates 159 bytes, likely related to control signaling or status reports. uRLLC traffic is concentrated in the range of 80–159 bytes, reflecting its reliance on small packet transmission for low

latency and high reliability. Internet traffic shows a more diverse pattern of distribution, with concentrations in both the 80–159 byte and 1280–1600 byte ranges, indicating a mix of small and large packet transmissions. This diversity suggests the occurrence of both frequent small transmissions, such as TCP acknowledgments, and larger-scale data transfers, such as file downloads.

The above-mentioned results highlight the variations in network demands between different applications, providing valuable insights into classifying traffic management in 5G networks.

B. Spatial Attention Enhanced CNN: SAE-CNN

A close relationship exists between the macro-level statistical characteristics of different types of network traffic and the micro-level packet features. Besides, this relationship can be effectively captured using deep learning models. Traffic types, such as eMBB, uRLLC, and Internet traffic, exhibit distinct statistical patterns, including differences in packet size, payload structures, and byte sequences. Manifested as distinct packet rates and frequency characteristics, these patterns are reflected in the distribution of traffic. The SAE-CNN is designed to extract spatial features from individual packets using convolution, automatically learning patterns in the byte sequences. Specifically, SAE-CNN captures peak byte distributions in eMBB traffic, stability features in uRLLC traffic, and irregularities in Internet traffic. Corresponding to the differences in packet count and traffic intensity as observed

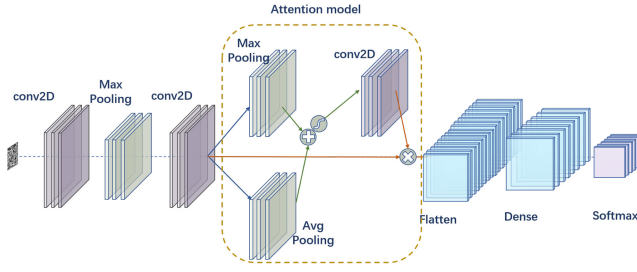


Fig. 8. Structure of proposed SAE-CNN model.

in macro-level distributions, these micro-level features enable the model to classify various traffic types effectively.

By combining convolutional networks with attention mechanisms, SAE-CNN can recognize fine-grained patterns in individual packets and classify traffic based on overall traffic distribution. This approach fully demonstrates the advantage of deep learning models in performing end-to-end analysis, from packet-level features to the overall traffic profile. Fig. 8 illustrates how the enhanced CNN is structured, with the relevant parameters listed in Table II. The SAE-CNN model consists of three convolutional layers, three pooling layers, and two fully connected layers. It takes preprocessed 32×32 grayscale images as input. The first convolutional layer extracts features using a 3×3 kernel, producing 32 feature maps, followed by an ReLU activation. A 2×2 max-pooling layer reduces the spatial dimensions to 16×16 . The second convolutional layer increases both the kernel size and the number of feature maps to 64 , followed by another max-pooling layer, reducing the size to 8×8 . A Dropout layer with a rate of 0.5 is applied to reduce overfitting. Two Lambda layers perform 5×5 average pooling, generating two 16×16 feature descriptors, which are concatenated into a $16 \times 16 \times 2$ feature map. A 3×3 convolution generates a spatial attention map, which is applied to the feature map through element-wise multiplication. Finally, the features are flattened and passed through two fully connected layers for classification. One of the core innovations of SAE-CNN lies in the introduction of the spatial attention mechanism, which enables the model to better focus on specific spatial locations by globally modeling the spatial dimensions of input features. To achieve this, the convolutional block attention module (CBAM) module proposed by Woo et al. [43] is simplified, focusing solely on spatial attention. The network performs dual average pooling operations on the input feature maps, aggregating the spatial information. Experiments show that dual average pooling retains global spatial information better than max pooling. The spatial attention map is computed as follows:

$$\begin{aligned} M_s(F) &= \sigma \left(f^{5 \times 5} ([AvgPool(F); AvgPool(F)]) \right) \\ &= \sigma \left(f^{5 \times 5} \left(\begin{bmatrix} F_{avg}^s \\ F_{avg}^s \end{bmatrix} \right) \right) \end{aligned} \quad (1)$$

where σ denotes the Sigmoid function, $f^{5 \times 5}$ represents a 5×5 convolution operation, and F_{avg}^s represents the feature maps generated by average pooling. The spatial attention map enhances the model's ability to capture critical positional features by encoding the weights of different spatial locations.

TABLE II
DETAILED STRUCTURE OF PROPOSED SAE-CNN MODEL

Layer	Type	Filter	Output
1	Conv2D+Relu	3×3	$32 \times 32 \times 32$
2	Max pooling	2×2	$16 \times 16 \times 32$
3	Conv2D+Relu	3×3	$14 \times 14 \times 64$
4	dropout(0.5)	-	-
5	Lambda_1	5×5	$16 \times 16 \times 1$
6	Lambda_2	5×5	$16 \times 16 \times 1$
7	concatenate	-	$16 \times 16 \times 2$
8	Conv2D	3×3	$16 \times 16 \times 1$
9	Mutifly	-	$16 \times 16 \times 64$
10	add	-	$16 \times 16 \times 64$
11	flatten	-	16384
12	Dense_1	-	64
13	Dense_2	-	3

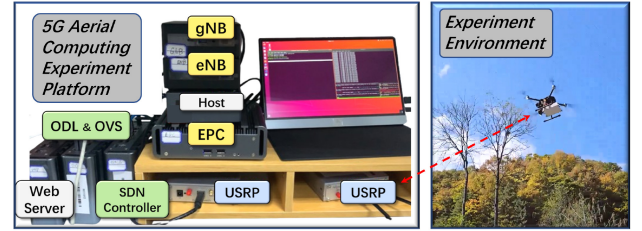


Fig. 9. Experimental platform and environment.

SAE-CNN, using this mechanism, improves detection of important byte positions in packets compared to traditional 2D-CNNs. This enhancement is particularly effective in traffic classification tasks, allowing the model to more accurately differentiate between traffic types, such as eMBB, uRLLC, and Internet traffic.

V. EXPERIMENTS ANALYSIS

In this section, three experiments are conducted to validate the performance of the proposed model. First, the effectiveness of the SAE-CNN model is evaluated on the collected UAV traffic dataset, and the impact of different input image sizes (32×32 , 36×36 , and 40×40) on classification performance is examined. A comparison with the performance of the 2D-CNN model before improvements is also made. Then, comparative experiments are carried out using classical CNNs and similar methods based on the open-source ISCXTor2016 dataset, with metrics, such as accuracy, cross-entropy loss, recall, and F1 score being evaluated. Lastly, a 5G heterogeneous network business-aware platform is built to validate the SAE-CNN models' feasibility in actual network environments.

Fig. 9 shows the experimental platform and environment. A 5G-NSA network has been successfully constructed using OpenAirInterface,^{1,2} adhering strictly to the official guidelines

¹Openairinterface5G, [online] Available: <https://openairinterface.org/oai-5g-ran-project/>.

²Oai.github, [online] Available: <https://gitlab.eurecom.fr/oai/openairinterface5g>.

and procedures. In the UAV experimental setting, the 5G experiment platform is built using the NUC11PAH model, which features an Intel Core i5-1135G7 processor, 8GB of RAM, and 256GB of storage. The software environment includes Ubuntu 18.04, OpenAirInterface 5G development tools, Python 3.7.3, and Docker 18.06. A mini-PC with an Intel J4125 processor, 8GB of RAM, and 512GB of storage is utilized for the OVS and the SDN controller, which is configured with Nginx, Docker, ODL, JDK 1.8, and Mininet for network simulation. All devices operate on Ubuntu 18.04 and are configured according to specific application requirements. The experimental setup maintains a pristine electromagnetic environment, free from interference. The UAV operates at altitudes ranging from 50 to 100 m, while the distance from the ground control station is kept within 100 to 200 m. This configuration ensures a stable and reliable communication link, which is essential for the operation of the UAV.

A. Evaluation Metrics

This study employs standard performance metrics commonly used in classification experiments, including cross-entropy loss, accuracy, precision, recall, and F1 score. In addition to these, important metrics, such as model runtime and model size are also considered to assess the model's feasibility for real-world applications. The specific formulas for these metrics are as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (2)$$

$$\text{Cross-Entropy Loss} = - \sum_{i=1}^N y_i \log(p_i) \quad (3)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{F1 Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

where TP is true positive, TN is true negative, FN is false negative, and FP is false positive.

B. Data Preprocessing

To ensure that the neural networks are trained on balanced datasets and to address issues related to class imbalance, equal representation of data for eMBB, URLLC, and Internet packet types is maintained. Since the Internet packet type contains the fewest packets (20 000 in total), it is selected as the baseline for analysis. For consistency, the first 20 000 packets are extracted from both the eMBB and URLLC datasets. During the data preprocessing phase, the payloads from the first 20 000 packets of each type are processed, with the first 1024, 1296, or 1600 bytes of each payload either extracted or zero-padded to create pixel-sized data byte streams. These streams are then preprocessed and converted into $N \times N$ (where N is 32, 36, or 40) pixel images, resulting in a total of 20 000 images. The images are randomly split into a training set and a test set, following a 9:1 ratio.

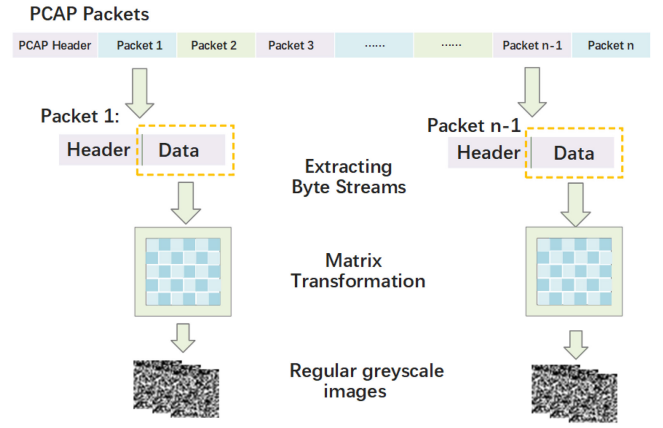


Fig. 10. Data preprocessing process.

A 2-D CNN is employed for classification. According to the properties of CNNs, the collected user data packets are converted into regular 2-D images, as shown in Fig. 10. First, the data portion of each packet is extracted based on the PCAP packet structure. Then, the extracted data are converted into a regular 2-D matrix, according to the size of the output image. After obtaining the 2-D matrix, data augmentation techniques, such as rotation, flipping, and deformation, are applied. Finally, the processed 2-D matrix is saved in JPEG image format.

To obtain packet data, the packet length is first determined as described in Section IV-A1). The packet data are then extracted as P_d . After extracting the packet data, a 2-D matrix conversion is required. That is, according to the predetermined image size ($n \times n$), the extracted data are first processed with a uniform length. The specific formula is as follows:

$$L = \text{len}(P_d), \quad (7)$$

$$\text{Processed Data} = \begin{cases} P_d + \text{ZP}(n^2 - L), & \text{if } L < n^2 \\ P_d[0 : n^2], & \text{if } L \geq n^2 \end{cases} \quad (8)$$

where ZP denotes zero padding.

Equation (8) evaluates the length L of the Packet Data and decides whether to apply zero padding (if the length is less than n^2) or to truncate the data (if the length is greater than or equal to n^2), ensuring that the processed data fits perfectly into a $n \times n$ matrix for further analysis or image conversion. After standardizing the data length, it can be reshaped into a 2-D matrix $\text{Matrix}_{n \times n}$

$$\text{Matrix}_{n \times n} = \text{reshape}(\text{Processed Data}, (n, n)). \quad (9)$$

At this point, the data has been extracted, and it is sufficient to save it as a grayscale image.

C. Model Evaluation on 5G Aerial Computing Dataset

In this section, the performance of the proposed SAE-CNN model is evaluated using the 5G aerial computing Dataset. This evaluation includes a detailed analysis of image preprocessing, feature extraction, model comparison, and performance metrics.

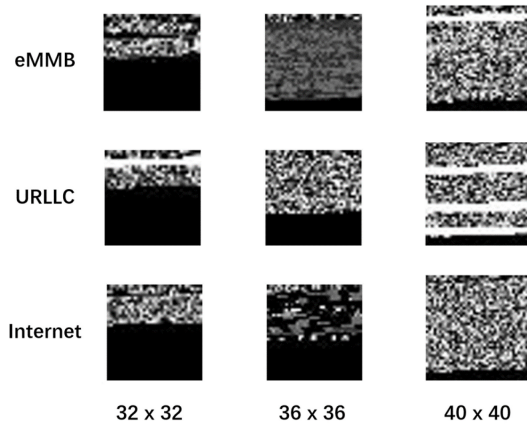


Fig. 11. Details of processed data.

TABLE III
COMPARATIVE PERFORMANCE OF SPATIAL FEATURE EXTRACTION
TECHNIQUES IN TRAFFIC CLASSIFICATION

Algorithm	Avg+Max	Avg+Avg	Max+Max
Accuracy	0.980	0.987	0.979
Loss	0.0124	0.0184	0.0135
Recall-eMBB	0.96	0.95	0.96
Recall-URLLC	0.98	0.98	0.97
Recall-Internet	0.96	0.98	0.95

1) *Feature Extraction With Spatial Attention*: Fig. 11 shows the grayscale image after data preprocessing. It is observed that the effective information in each grayscale image accounts for only about half. Therefore, the network is designed to focus on extracting features from the upper half of the image, which proves to be more effective. Inspired by the work in [43], the proposed CBAM self-attention mechanism is simplified, and the spatial interrelationships of the features in the network structure are used to generate spatial attention maps, enhancing the CNN's ability to extract features. It is found that using continuous average pooling twice is more suitable for the experimental environment, as shown by the results in Table III. Various pooling strategies Avg+Max, Avg+Avg, and Max+Max are tested for spatial feature extraction. The Avg+Avg pooling method achieves the highest accuracy (0.987) and lowest loss (0.0184), confirming that continuous average pooling effectively captures the necessary spatial relationships within the data for this experimental environment.

2) *Resolution Analysis*: The effect of different input feature resolutions (32×32 , 36×36 , and 40×40) on the performance of the SAE-CNN and a baseline 2D-CNN model is explored. In deep learning, input resolution plays a critical role in determining the amount of information the model can extract from each feature map. Higher resolutions provide more detailed information, which may improve classification accuracy, but also increase computational load and inference time. Conversely, lower resolutions reduce the computational burden but may lead to information loss, affecting the model's ability

TABLE IV
IMPACT OF DIFFERENT INPUTS ON THE PERFORMANCE OF SAE-CNN
AND 2D-CNN MODELS

Model	Image Size	Loss	Accuracy	Recall (Avg)	F1 Score (Avg)
2D-CNN	32	0.1325	0.9710	0.97	0.97
	36	0.1184	0.9778	0.98	0.98
	40	0.7067	0.7808	0.79	0.78
SAE-CNN	32	0.0325	0.9755	0.97	0.96
	36	0.0184	0.9842	0.98	0.99
	40	0.358	0.8950	0.90	0.89

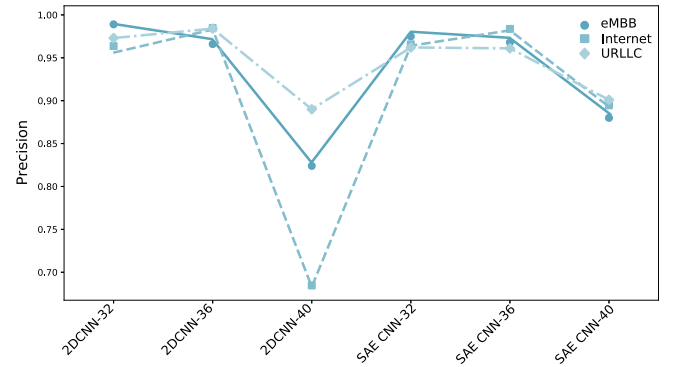


Fig. 12. Comparison of precision between SAE-CNN and 2D-CNN with different inputs feature resolution.

to distinguish complex patterns within the data. In the context of 5G UAV real-time traffic classification, finding an optimal balance between resolution and computational efficiency is essential, as UAVs typically have limited processing power and operate in scenarios requiring rapid inference for timely decision-making.

The results, presented in Table IV, show that SAE-CNN consistently outperforms the 2D-CNN, especially at a feature resolution of 36×36 pixels, where a balance between computational efficiency and classification accuracy is achieved. This resolution provides enough spatial detail to capture critical traffic patterns while maintaining manageable computational demands, making it ideal for deployment in resource-constrained UAV systems that require fast and accurate classifications.

Table IV and Fig. 12 provide a comparative analysis of how different feature resolutions impact the performance of SAE-CNN and 2D-CNN. While 2D-CNN shows strong performance at smaller resolutions (32×32 and 36×36), its accuracy declines at larger sizes (40×40). In contrast, SAE-CNN maintains stable accuracy across all resolutions, with optimal results observed at 36×36 . This suggests that SAE-CNN is more robust to variations in feature resolution. Overall, SAE-CNN demonstrates superior stability and accuracy across service types (eMBB, URLLC, and Internet) and consistently outperforms 2D-CNN, establishing 36×36 as the optimal input resolution for the model.

3) *Comparison With Baseline Models*: This experiment addresses two key challenges in UAV 5G traffic classification:

TABLE V
COMPARATIVE WITH BASELINE MODELS

Model	Loss	Accuracy	Recall (Avg)	F1 Score (Avg)
SAE-CNN	0.0184	0.9842	0.98	0.99
Alexnet	0.0487	0.9604	0.95	0.97
Resnet	0.0242	0.9720	0.97	0.96
DenseNet121	0.0163	0.9516	0.94	0.95

1) enhancing accuracy across diverse traffic types and 2) improving robustness in handling complex, overlapping data characteristics. As demonstrated in Section IV-A, the UAV traffic dataset exhibits significant complexity due to variations in statistical frequency, packet length, and overlapping traffic, which complicates accurate classification across service types.

The SAE-CNN model incorporates a spatial attention mechanism to improve feature extraction, allowing the model to focus on essential byte positions within packet sequences. This mechanism enhances SAE-CNN's ability to isolate features specific to each traffic type, reducing confusion between overlapping categories. This is particularly advantageous in distinguishing eMBB and URLLC traffic, where patterns may overlap but require precise categorization due to differing QoS requirements.

In comparison with traditional models, such as 2D-CNN, SAE-CNN achieves superior results, with an average accuracy of 98.42% and an F1 score of 0.99. These results demonstrate the model's enhanced classification capability for 5G aerial computing tasks. Based on the findings, a 36×36 pixel image is identified as the optimal input size for the model. Additional comparisons with deep learning networks (e.g., AlexNet, ResNet, and DenseNet121) show SAE-CNN achieving the best performance across loss, accuracy, recall (mean), and F1 score (mean), with accuracy reaching 98.42% and F1 score 0.99, as shown in Table V.

Figs. 13–15 illustrate the precision, recall, and F1 scores for baseline models across various service types. SAE-CNN consistently performs best across metrics for eMBB services, performs comparably to AlexNet in Internet services, and outperforms all models in URLLC services. Overall, SAE-CNN achieves scores of 99% and above across metrics, confirming its effectiveness in 5G UAV traffic classification.

D. Model Evaluation on ISCX Dataset

This experiment evaluates the performance of the proposed SAE-CNN model on the ISCX TOR 2016 dataset to address two main challenges in traffic classification: 1) enhancing accuracy across diverse traffic types and 2) reducing inference latency for real-time applications. The dataset includes eight traffic categories: browsing, audio, video, chat, email, VoIP, FTP, and P2P. The ISCX TOR 2016 dataset presents substantial complexity, with diverse protocols leading to imbalanced feature distributions, and significant variations in packet size, transmission frequency, and timing patterns. These factors

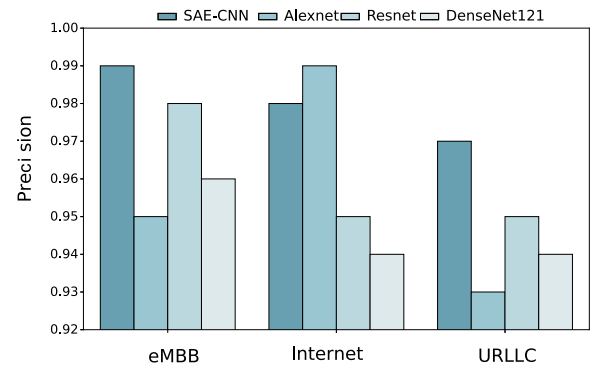


Fig. 13. Comparison of precision between SAE-CNN with baseline models.

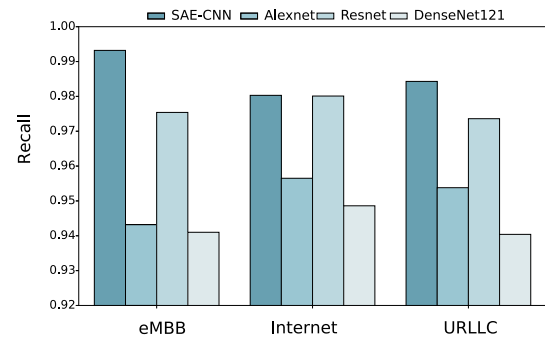


Fig. 14. Comparison of recall between SAE-CNN with baseline models.

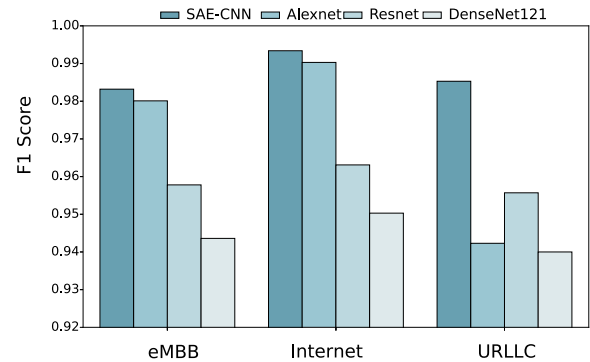


Fig. 15. Comparison of F1 Score between SAE-CNN with baseline models.

complicate classification by increasing overlap in statistical features among traffic types. Additionally, the dataset's imbalanced nature and presence of anomalies (e.g., unusual packet sizes, and burst patterns) highlight the need for models robust to fluctuations in network conditions.

Guided by recent studies [35], [36], [37], a balanced selection of traditional and advanced architectures is used to comprehensively evaluate SAE-CNN's performance. Comparisons are made with SOTA models like AlexNet, ResNet50, and DenseNet121, along with recent models, such as 1D-CNN and LSTM, widely applied in network traffic classification. All models are trained using stochastic gradient descent (SGD) with a learning rate of 0.001, a batch size of 32, and 20 training epochs. Table VI and Figs. 16–18 present a comparative analysis of SAE-CNN and SOTA methods on the ISCX dataset. SAE-CNN achieves the highest accuracy of 98.72%, marking an improvement of 0.92% to 1.42% over

TABLE VI
COMPARISON BETWEEN SAE-CNN WITH SOTA METHODS BASED ON ISCX DATASET

Model	Loss	Accuracy	Precision(Avg)	Recall(Avg)	F1 Score(Avg)	Total Inference Time
SAE-CNN	0.0104	0.9872	0.9951	0.9821	0.9975	58.2s
LSTM	0.102	0.978	0.9787	0.9812	0.9712	352 s
1D-CNN	0.131	0.973	0.9731	0.9725	0.9737	472s
Alexnet	0.0077	0.9851	0.9832	0.9856	0.9723	194.6 s
Resnet50	0.0126	0.9763	0.9715	0.9738	0.9623	269.5 s
DenseNet121	0.0726	0.9497	0.9531	0.9541	0.9482	274.3 s

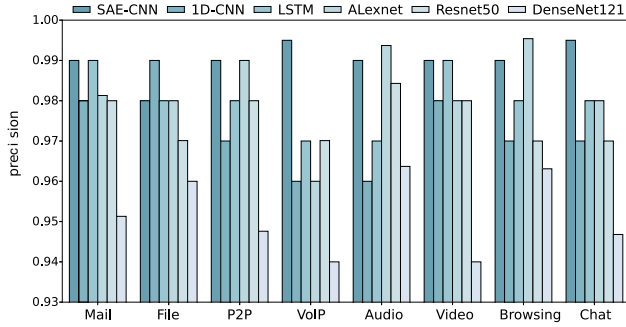


Fig. 16. Comparison of precision between SAE-CNN with SOTA methods on ISCX Dataset.

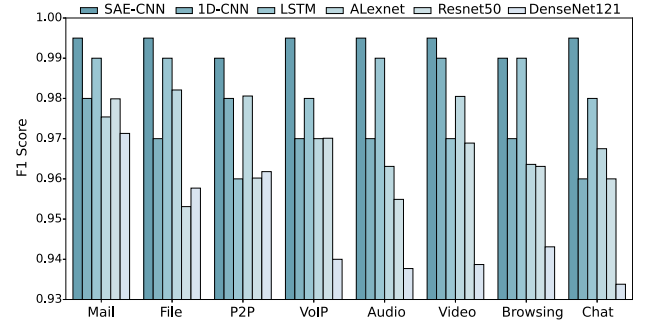


Fig. 18. Comparison of F1 Score between SAE-CNN with SOTA methods on ISCX Dataset.

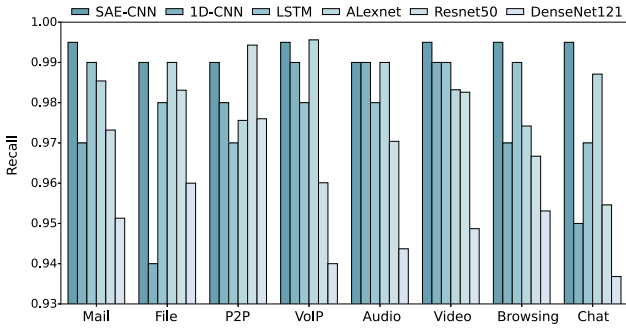


Fig. 17. Comparison of recall between SAE-CNN with SOTA methods on ISCX Dataset.

other models, demonstrating superior classification capability in complex scenarios. With a precision of 99.51% and an F1 score of 99.75%, SAE-CNN effectively distinguishes between overlapping traffic types, such as browsing, chat, and video streaming. The spatial attention mechanism enhances classification accuracy and reduces latency by focusing on critical byte positions, allowing SAE-CNN to capture unique traffic patterns and address protocol diversity and temporal variability. One of SAE-CNN's key advantages is the total inference speed. With an average inference time of 58.2 s, it significantly outperforms models like LSTM (352 s) and 1D-CNN (472 s), making it ideal for real-time applications in 5G networks where low latency is crucial. Additionally, SAE-CNN's model size is only 0.847 MB, ensuring high memory efficiency.

These results underscore SAE-CNN's suitability for real-world applications, where both performance and efficiency are essential. The reduced inference time and compact model size

make SAE-CNN particularly advantageous for deployment in environments like UAV networks or edge computing, where computational resources and processing speed are limited.

E. Feasibility Validation

To demonstrate the feasibility of the proposed SAE-CNN method for traffic detection and classification in real network environments, a 5G heterogeneous network management platform is developed using the open-source D2Admin framework.³ The platform integrates modules for network topology display, state detection of base stations and users, and service awareness. Fig. 19 illustrates the platform's interface, which shows the network topology, device labels, and connections, with four interactive buttons for various functions.

The business-aware module, built using the Flask framework, encapsulates the SAE-CNN classifier and allows communication with the front-end. Upon clicking the Service Type Awareness button, a GET request is sent to the SAE-CNN backend. The backend captures network data using Wireshark, preprocesses it to match the training data format, and feeds the data into the pretrained SAE-CNN classifier for classification.

The data preprocessing involves traffic slicing, extracting byte streams, converting the data to grayscale images, and assigning business labels, before passing it to the SAE-CNN model. After classification, the results are encapsulated in JSON format and returned to the front-end, where they are parsed and displayed to the user via a pop-up window. For testing, a UAV connected to the 5G network is used to request live streaming. The base station accesses the platform to initiate service perception, and the business-aware results are shown in Fig. 19. Part 3 of the figure displays the platform's

³D2Admin, [online] Available: <https://github.com/d2-projects/d2-admin>.

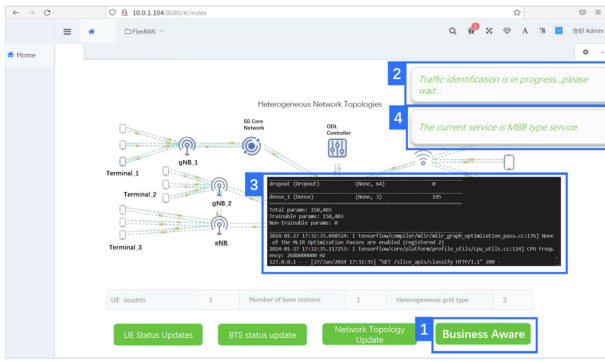


Fig. 19. Business-aware results on 5G heterogeneous network management platform.

runtime log output, and part 4 shows the business-aware classification results.

VI. CONCLUSION

UAV aerial computing applications are highly compatible with the high-bandwidth and low-latency capabilities of 5G networks. However, research on 5G and UAVs, particularly in complex traffic classification, remains limited. This paper presents a 5G-enabled UAV traffic classification framework that leverages SDN technology to support heterogeneous communication mechanisms, such as LTE and 5G NSA. Three major categories of 5G aerial computing service traffic data are collected to evaluate the proposed approach. A novel traffic classification model, SAE-CNN, is introduced, incorporating spatial attention mechanisms to significantly improve accuracy and reduce latency in UAV traffic classification. The model demonstrates superior performance across various 5G service scenarios, outperforming SOTA methods. Extensive experimental validation highlights the robustness of the SAE-CNN model across multiple 5G service types, including eMBB, URLLC, and Internet traffic. Additionally, a 5G heterogeneous network management platform is implemented to verify the real-world feasibility of the proposed model. The platform successfully integrates the SAE-CNN classifier, enabling real-time traffic detection and management, which is crucial for UAV operations in dynamic and complex 5G environments.

The UAV traffic dataset, SAE-CNN model, and deployment framework presented in this study provide a solid foundation for future research on resource allocation and QoS assurance within 5G UAV networks. By enabling efficient traffic classification, this approach supports critical applications, such as emergency response, smart city management, and environmental monitoring, where precise traffic management and reliable communication are essential. The capabilities of this method thus hold substantial potential to advance both practical applications and ongoing research in these areas.

Although the proposed model demonstrates significant advancements, it has not yet been evaluated under highly dynamic conditions or unpredictable settings, such as areas prone to severe and frequent signal interference. Consequently, challenges remain in service identification within 5G UAV networks, including handling complex traffic patterns, addressing weak signal conditions in disaster relief scenarios, and

managing signal interference in urban environments. These factors affect the performance and reliability of UAV operations and traffic classification. Additionally, selecting the optimal deployment location for traffic classification systems in 5G networks is critical. Deploying at the core network allows for comprehensive traffic analysis but may increase latency, impacting real-time UAV services. Alternatively, placing the system at the access layer reduces latency for critical UAV applications but may limit traffic analysis depth, affecting traffic management efficiency.

For future work, further data collection and analysis of UAV transmission signals in real-world scenarios, such as disaster relief operations under weak signal conditions and urban management tasks amid signal interference are planned to validate performance across diverse 5G UAV environments [44], [45]. The SAE-CNN model will be tested on newer 5G datasets to evaluate its performance across a broader range of real-world conditions, providing deeper insights into the model's generalizability and applicability to evolving traffic patterns [46]. Moreover, the integration of SAE-CNN with advanced 5G networks, such as network slicing, will be explored to enhance efficiency and adaptability in managing resource-constrained UAV networks. Regarding deployment challenges, future research should focus on developing adaptive traffic classification solutions that dynamically determine the optimal deployment location based on real-time network conditions and traffic patterns [47]. Additionally, more input features should be considered by combining physical layer signal features with the deep-residual-learning-based cognitive model [48] to enhance network performance and adaptability in dynamic 5G UAV environments.

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